Constrained static hedging of a capped GBM claim

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Abstract

I study static hedging of a capped European claim under geometric Brownian motion with position and funding limits. Risk-neutral prices determine the initial premium and instrument costs, while the physical measure determines mean-square discounted terminal error. Dividends are reinvested explicitly. Closed truncated-lognormal moments yield a strictly convex quadratic loss and a unique constrained hedge. Complete optimality proofs include the zero-budget boundary, a finite active-set method, the distinction between variance and mean-square objectives, and budget shadow prices. I compare 126 constrained hedges with unrestricted and cash-only baselines, a same-constraint variance optimiser, a projection heuristic and an independent general-purpose solver. Monte Carlo and separate residual-square integration check the risk calculations. Adverse heuristic rankings, small numerical violations and unresolved cancellation cases remain visible. Funding limits have an exact risk penalty; additional budget has a locally interpretable marginal value. The study provides a reproducible financial optimisation example with rigorous proofs, using synthetic inputs and established hedging foundations.

Keywords: static hedging; mean-square loss; physical measure; convex optimisation; funding constraints; KKT conditions; shadow prices.

JEL classification: C61, C63, G13.

1 Financial model and funded loss

I study how position and funding constraints alter static hedging of a capped claim under a classical geometric Brownian motion (GBM) model. Static quadratic hedging with pricing-measure cost constraints is established (Leung and Lorig, 2016); I specialise it to a two-instrument basis and a fixed no-arbitrage premium. Fixed-capital mean-square loss also appears in constrained dynamic hedging (Czichowsky and Schweizer, 2012), but the existence proof here is finite-dimensional. [1] The underlying GBM market remains dynamically complete. Residual risk arises from the restricted instrument set and static positions. Physical parameters are stipulated for synthetic comparisons.

Let $S, K, H, T, \sigma > 0$, let r, q, μ be finite constants, and set $J = K + H$, $D = e^{-rT}$ and $a = e^{(q-r)T}$. Under the physical measure,

$$X = S\exp\{(\mu - q - \sigma^2/2)T + \sigma\sqrt{T}Z\}, \qquad Z \sim N(0,1), \qquad Y = D\min\{(X-K)^+, H\}. \quad (1)$$

Here μ is the total-return drift. I price the contract under the stipulated equivalent measure Q, whose ex-dividend drift is $r - q$. For strike A, let

$$\begin{aligned} C_A &= Se^{-qT}\Phi(d_{1,A}) - AD\Phi(d_{2,A}), \\ d_{1,A} &= \frac{\log(S/A) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_{2,A} = d_{1,A} - \sigma\sqrt{T}. \end{aligned} \quad (2)$$

[1] The closest static antecedent was read in its identified author preprint; its published successor metadata was verified. The source ledger records that distinction and the limited sections actually consulted.

These classical prices follow Black and Scholes (1973) and Merton (1973). The received premium is $V_0 = C_K - C_J > 0$. Under Q, terminal stock has a strictly positive lognormal density on $(0, \infty)$. The non-negative call and capped payoffs are positive on sets of positive probability, so their finite discounted expectations give $C_K > 0$ and $V_0 > 0$. I allow only the total-return stock, with dividends reinvested, and the vanilla call at K. Their initial costs and discounted gains are

$$p = (S, C_K)^\top, \qquad g = (aX - S,\ D(X-K)^+ - C_K)^\top. \tag{3}$$

For fixed holdings $\theta = (\theta_s, \theta_c)^\top$, the initial bank balance is $V_0 - p^\top\theta$. The self-financing portfolio's discounted terminal wealth is $V_0 + \theta^\top g$, so the discounted liability residual and loss are

$$R(\theta) = Y - V_0 - \theta^\top g, \qquad L(\theta) = \mathbb{E}_P[R(\theta)^2] = c - 2h^\top\theta + \theta^\top M\theta, \tag{4}$$

where $M = \mathbb{E}_P[gg^\top]$, $h = \mathbb{E}_P[g(Y - V_0)]$ and $c = \mathbb{E}_P[(Y - V_0)^2]$. This is uncentred mean-square financial loss, including squared bias, rather than residual variance alone. It penalises discounted terminal surplus and shortfall symmetrically; the pricing premium is fixed, not chosen to minimise this loss. I impose the declared box and instrument-spend budget

$$\mathcal{F}_B = \{\theta : 0 \le \theta_s, \theta_c \le 1,\ p^\top\theta \le B\}, \qquad B = \kappa V_0, \quad \kappa \ge 0. \tag{5}$$

For $\kappa \le 1$ the bank balance is non-negative; for $\kappa > 1$ borrowing may occur. These position bounds are modelling choices, not universal limits on optimal hedges.

2 Convexity and constrained optimality

The relevant convex optimality principles are established (Boyd and Vandenberghe, 2004). The following specialisation includes the zero-budget boundary, where strict feasibility fails.

Theorem 2.1 (Existence, uniqueness and optimality)**.** *The matrix M is positive definite and $\mathcal{F}_B$ is nonempty and compact. The constrained hedge is unique. Its necessary and sufficient conditions are feasibility, non-negative multipliers $\ell, u \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$, and*

$$2(M\theta - h) - \ell + u + \lambda p = 0, \tag{6}$$

$$\ell_i\theta_i = 0, \quad u_i(\theta_i - 1) = 0, \quad \lambda(p^\top\theta - B) = 0. \tag{7}$$

These conditions also hold at $B = 0$, although strict feasibility fails there. The optimum lies among at most sixteen candidates: the unconstrained stationary point, stationary points on five supporting lines and their ten pairwise intersections, retaining only well-defined feasible candidates.

Appendix A gives full proofs and closed Gaussian moment expressions for every objective coefficient. It also expresses risk as an integral of three squared affine residuals. No finite hedge in this instrument span replicates the capped claim exactly; apparent zero risk from subtracting nearly equal floating-point moments is therefore a numerical warning. A long call at K and short call at J would replicate the claim, but that second call is unavailable here.

3 Loss choice and the value of funding

3.1 Variance minimisation and mean-square hedging

The financial objective also explains why variance-reduction controls cannot simply be reused as optimal hedges. Put $m = \mathbb{E}_P g$, $e = \mathbb{E}_P(Y - V_0)$, $G = \mathrm{Cov}_P(g)$ and $k = \mathrm{Cov}_P(g, Y - V_0)$. With $\theta_v = G^{-1}k$, $\zeta = m^\top G^{-1}m$ and $\eta = e - m^\top\theta_v$, the unrestricted mean-square relaxation is

$$\theta_M = M^{-1}h = \theta_v + \frac{G^{-1}m\,\eta}{1+\zeta}, \qquad L(\theta_v) - L(\theta_M) = \frac{\eta^2\zeta}{1+\zeta}. \tag{8}$$

Contract	S	K	H	T	r	q	σ
Lower kink	100	100	20	1	0.03	0.01	0.2
Upper kink	120	100	20	1	0.03	0.01	0.2
Inside spread	110	100	20	1	0.03	0.01	0.2
Deep cap	160	100	20	1	0.03	0.01	0.2
Short dated	100	100	20	0.05	0.03	0.01	0.06
Negative rate/dividend	105	100	20	1	-0.015	0.045	0.25
High volatility	95	100	30	2	0.03	0.01	0.65

Table 1: Designed one-asset inputs. Instrument costs and the received premium use Q; terminal loss uses the specified physical drift. No input is a market estimate.

Under $P = Q$, $m = e = 0$ and the objectives coincide; different measures permit, but do not force, a strict difference. This unrestricted L^2 benchmark can contain short positions and is not automatically admissible under the bounded-below wealth condition. Constraints add the exact penalty $L(\theta_B) - L(\theta_M) = (\theta_B - \theta_M)^\top M(\theta_B - \theta_M)$.

3.2 Budget shadow prices

The optimal risk $v(B) = \min_{\mathcal{F}_B} L$ is non-increasing and convex. Every budget multiplier yields the global supporting bound

$$v(B') \geq v(B) - \lambda_B(B' - B), \qquad B' \geq 0. \tag{9}$$

On a regular active-set interval, $v'(B) = -\lambda_B$ and v is quadratic or constant. At zero budget, $v'(0+) = -2\max\{0, h_s/S, h_c/C_K\}$: the initial marginal benefit of available budget is the largest positive loss-reduction/cost ratio; unhelpful spending is not required. The budget here permits spending; it does not require the entire amount to be invested.

4 Registered comparisons and numerical evidence

I retain the seven designed contracts in Table 1. The protocol was saved before outcomes, with physical total-return drifts $\mu \in \{0.03, 0.06, 0.12\}$ and budget multiples $\kappa \in \{0, 0.25, 0.5, 1, 2, 4\}$. This gives 126 constrained mean-square comparisons. Rates are annualised; maturities are in years. The parameters are synthetic, with no licensed or empirical observations. The experiment assumes a known physical law; it does not fit μ.

The analytic unconstrained solution is the strongest two-instrument relaxation, and cash-only financing is the no-hedge baseline. I compare the constrained solution with a separately solved centred-variance objective using the same constraints, a Euclidean projection heuristic, and SciPy's general-purpose SLSQP optimiser. The projection heuristic minimises distance from unrestricted variance-optimal holdings; it does not minimise financial loss. The exact call-spread reference uses an additional call and a short position, so it identifies the importance of the instrument restriction rather than superiority within a shared feasible set.

Closed truncated moments determine objective coefficients. Independent adaptive normal integration, split at both payoff kinks, checks 168 mean/product coefficients. Risk is reported primarily by integrating the non-negative squared residual, with the expanded moment quadratic retained alongside it. I do not replace negative computed risks with zero or repair eigenvalues. The numerical mean-square matrix condition number reaches $42,337$, despite exact positive definiteness. Thus numerical representation remains a separate issue from the convexity proof.

No registered optimiser failure or quadrature warning occurred. Maximum scaled primal violation is 1.88×10^{-14}; maximum negative multiplier magnitude is 2.84×10^{-14}. The largest stationarity and complementary-slackness residuals are 4.55×10^{-13} and 2.76×10^{-15} respectively.

Contract	No hedge	Constrained	Unrestricted	Reduction (%)
Lower kink	68.4032	15.7205	13.5707	77.02
Upper kink	58.6954	24.9738	20.2983	57.45
Inside spread	71.9973	22.9439	19.0941	68.13
Deep cap	6.06651	5.23772	4.218	13.66
Short dated	0.770884	Unresolved	Unresolved	n/a
Negative rate/dividend	85.0105	24.4634	20.1739	71.22
High volatility	151.748	77.0938	57.1103	49.20

Table 2: Population-risk diagnostics at $\mu = 0.06$ and $B = V_0$. Losses are squared discounted monetary units. The percentage compares the constrained mean-square hedge with cash-only financing. Unrestricted holdings relax the box and budget. The short-dated hedge risks lie below a heuristic arithmetic floor and support neither a resolved risk ratio nor a zero-risk assertion.

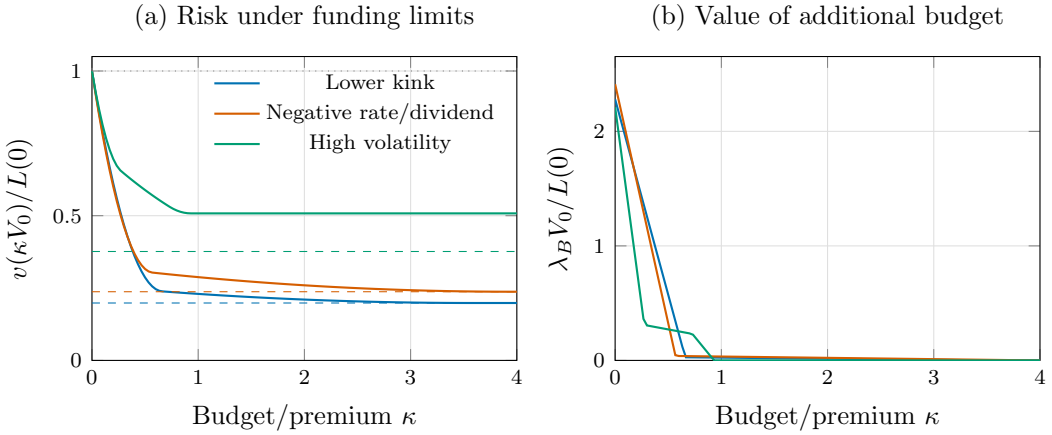


Figure 1: Funding and residual risk for three registered contractual scenarios at $\mu = 0.06$. Solid curves use the constrained mean-square hedge; dashed horizontal levels give each two-instrument unrestricted relaxation. The dotted level is the cash-only baseline. The right panel shows the normalised budget multiplier, equal to the marginal reduction in normalised risk on regular active-set intervals. At zero I use the proved right derivative. A supplemental 121-point grid illustrates curvature and saturation; all seven contracts remain in the primary comparison. These are synthetic floating-point diagnostics.

These are floating-point diagnostics, not exact feasibility or admissibility certificates. SLSQP's residual-square risks differ from the active-set values by between -2.27×10^{-13} and 1.59×10^{-11}, with maximum position difference 2.34×10^{-7}. The small negative comparison remains visible. Equation (23) states the exact prerequisites needed to translate residuals into a risk bound.

Among the 117 constrained rows above the heuristic floor, reductions against the no-hedge baseline range from zero to 81.5868 per cent. The largest same-constraint variance-objective penalty is 0.191175 squared monetary units, or 0.5982 per cent of the mean-square optimum, for the negative-rate/dividend contract at $\mu = 0.12$, $\kappa = 0.5$. The distinction is mathematically substantive but quantitatively modest in this registered family. Euclidean projection has a largest penalty of 612.244, illustrating why geometric closeness to an unconstrained holding is not a financial risk criterion.

Figure 1 links the funding constraint to both residual risk and its local marginal value. I use two-dimensional curves because each panel varies one budget parameter. The supplemental grid adds resolution to the function, not further selected hedge trials. The primary tables retain every registered contract and physical law.

I also evaluate all holdings on 65,536 independent normal paths per contract/physical-law

regime, using seed 674193 plus $1009(3i+j)$ for zero-based contract and drift indices. The same paths compare the methods within a regime, yielding 420 Monte Carlo rows. No path is used to fit a hedge and no raw paths are retained. The reported standard errors concern sample means of squared residuals and are asymptotic diagnostics. All residual fourth moments are finite under the stipulated lognormal law, supporting the usual independent-sample convergence assumptions; these standard errors are not finite-sample coverage certificates. The maximum absolute standardised discrepancy is 2.21075. Two rows exceed 1.96, but they use identical holdings and the same paths, so they are not independent observations or simultaneous coverage evidence.

Three negative expanded-quadratic risks and nine heuristic-floor constrained rows remain in the archive. The floor is $128\epsilon_{\mathrm{mach}}$ times the sum of absolute quadratic terms; it is a diagnostic choice, not a proved roundoff bound. For the short-dated case the residual-square calculation gives values around 10^{-30}–10^{-28} for near-unit-call holdings, while moment subtraction can report zero or a negative number. The true two-instrument optimum is strictly positive by the proof. Numerical positivity alone does not resolve its magnitude: tiny coefficient/funding errors and rare unsampled tails can dominate these computed risks.

5 Interpretation and reproducibility

The financial question is how much discounted terminal error remains when a fixed fair premium must fund a restricted static hedge. The exact argument separates initial valuation under Q, physical loss under P, funding feasibility and numerical error. Its useful project-specific conclusions are the loss distinction, the exact constraint penalty and the interpretable budget frontier. These are checked specialisations of established hedging and convex optimisation, with reproducible adverse comparisons.

The engine is `gbm_hedging.py`. Its runner, `run_hedging_study.py`, reads `hedging_protocol.json` and writes a fresh result directory. Python 3.13, NumPy 2.3.5 and SciPy 1.16.3 reproduce the registered computations. Ten tests compare independent moment representations, active-set and general-purpose solutions, budget subgradients, dividend funding and cancellation cases. The source ledger records publication status, actual consulted versions and reading scope. Full derivations follow in the appendix. The existing pricing/calibration companion and main quantum paper retain their previous evidence.

A Moment, optimality and sensitivity proofs

A.1 Discounted financial accounting and exact moments

A fixed unit of the total-return stock has terminal value $e^{qT}X$; continuous dividends are reinvested, making the share count evolve mechanically as e^{qt}. A call pays $(X-K)^+$ and the bank balance accumulates at r. Thus discounted portfolio wealth is $a\theta_s X + D\theta_c(X-K)^+ + V_0 - p^\top\theta = V_0 + \theta^\top g$. For the constrained long-only positions, the available securities have non-negative prices. Discounted wealth is bounded below by the fixed initial bank balance, itself at least $V_0 - B$. This proves admissibility on the finite horizon even when borrowing occurs. The unrestricted L^2 relaxation need not satisfy that condition. Expanding the squared residual gives (4). All moments are finite because lognormal variables have every positive integer moment.

Write $b = \log S + (\mu - q - \sigma^2/2)T$, $v = \sigma\sqrt{T}$ and $t_j = \mathbb{E}_P X^j = \exp(jb + j^2v^2/2)$. For $j = 0, 1, 2$, define

$$\begin{aligned} T_j(l,u) &= \mathbb{E}_P[X^j \mathbf{1}_{\{l<X\le u\}}] \\ &= t_j\left[\Phi\left(\frac{\log u - b - jv^2}{v}\right) - \Phi\left(\frac{\log l - b - jv^2}{v}\right)\right]. \end{aligned} \tag{10}$$

The conventions $\log 0 = -\infty$ and $\log \infty = \infty$ cover tails. To derive this formula, integrate $e^{j(b+vz)}\varphi(z)$ over the corresponding normal interval. Completing the square, $jvz - z^2/2 = -(z - jv)^2/2 + j^2v^2/2$, gives the displayed Gaussian probability. Interval endpoints have zero probability.

Let $F = \min\{(X - K)^+, H\}$ and $A = (X - K)^+$. Put $f_0 = \mathbb{E}F$, $f_1 = \mathbb{E}XF$, $f_2 = \mathbb{E}F^2$ and $a_0 = \mathbb{E}A$, $a_1 = \mathbb{E}XA$, $a_2 = \mathbb{E}A^2$. Splitting at K, J gives

$$\begin{aligned}
f_0 &= T_1(K,J) - KT_0(K,J) + HT_0(J,\infty),\\
f_1 &= T_2(K,J) - KT_1(K,J) + HT_1(J,\infty),\\
f_2 &= T_2(K,J) - 2KT_1(K,J) + K^2T_0(K,J) + H^2T_0(J,\infty), \qquad (11)\\
a_0 &= T_1(K,\infty) - KT_0(K,\infty),\\
a_1 &= T_2(K,\infty) - KT_1(K,\infty),\\
a_2 &= T_2(K,\infty) - 2KT_1(K,\infty) + K^2T_0(K,\infty), \qquad (12)\\
z &= \mathbb{E}FA = T_2(K,J) - 2KT_1(K,J) + K^2T_0(K,J)\\
&\qquad + H\{T_1(J,\infty) - KT_0(J,\infty)\}. \qquad (13)
\end{aligned}$$

Expansion of the instrument gains now supplies

$$\begin{aligned}
m &= (at_1 - S,\ Da_0 - C_K)^\top,\\
M_{ss} &= a^2t_2 - 2aSt_1 + S^2,\\
M_{cc} &= D^2a_2 - 2DC_Ka_0 + C_K^2,\\
M_{sc} &= aDa_1 - aC_Kt_1 - SDa_0 + SC_K, \qquad (14)\\
h_s &= aDf_1 - SDf_0 - V_0(at_1 - S),\\
h_c &= D^2z - DC_Kf_0 - V_0(Da_0 - C_K),\\
c &= D^2f_2 - 2DV_0f_0 + V_0^2, \qquad e = Df_0 - V_0. \qquad (15)
\end{aligned}$$

There is a separate positive-integral representation. On the successive regions $(0, K)$, (K, J) and (J, ∞), write $R = A_iX + B_i$. The coefficients are

$$\begin{aligned}
(A_1, A_2, A_3) &= (-a\theta_s,\ D(1-\theta_c) - a\theta_s,\ -D\theta_c - a\theta_s),\\
B_1 &= -V_0 + S\theta_s + C_K\theta_c,\\
B_2 &= -DK(1-\theta_c) - V_0 + S\theta_s + C_K\theta_c,\\
B_3 &= DH + DK\theta_c - V_0 + S\theta_s + C_K\theta_c. \qquad (16)
\end{aligned}$$

Integrating $\sum_i \mathbf{1}_i(A_iX + B_i)^2$ against the positive lognormal density gives the same risk. This mathematical identity does not guarantee that either expanded floating-point calculation is free from cancellation.

A.2 Positive definiteness, existence and the risk penalty

The lognormal density is strictly positive on $(0, \infty)$. If $u^\top Mu = \mathbb{E}(u^\top g)^2 = 0$, continuity makes $u^\top g(x) = 0$ on that whole interval. Below K, its affine slope is au_s, so $u_s = 0$, and its constant is $-u_cC_K$, so $u_c = 0$ because $C_K > 0$. Thus $M \succ 0$. For $G = \mathrm{Cov}(g)$, zero quadratic variance would instead make $u^\top g(x)$ constant. The below-K slope again gives $u_s = 0$, while the above-K slope Du_c gives $u_c = 0$. Hence $G \succ 0$ as well.

The feasible set is closed, bounded and contains zero. Continuity therefore gives existence, and the positive-definite Hessian $2M$ gives strict convexity and uniqueness. Completing the quadratic yields

$$L(\theta) = c - h^\top M^{-1}h + (\theta - \theta_M)^\top M(\theta - \theta_M), \qquad \theta_M = M^{-1}h. \qquad (17)$$

This proves the constraint penalty stated in the main text. Nevertheless the unrestricted minimum is strictly positive. If any finite θ made $L = 0$, continuity and positive density would make all three affine residuals identically zero. Their slopes in (16) force, successively, $\theta_s = 0$, $\theta_c = 1$ and $\theta_c = 0$, a contradiction. The unrestricted minimum is attained at θ_M, so this is a strictly positive minimum, not merely positivity at each chosen hedge.

A.3 KKT necessity without a constraint qualification

Write the five inequalities as $A\theta \leq b(B)$, with rows $-e_s^\top, -e_c^\top, e_s^\top, e_c^\top, p^\top$ and right sides $0, 0, 1, 1, B$. At a minimiser θ_*, any direction d with $A_I d \leq 0$ for the active rows I remains feasible for all sufficiently small positive steps: each inactive inequality has a positive slack. Differentiability gives $\nabla L(\theta_*)^\top d \geq 0$. For completeness, let K_I be the cone of the active row normals. This cone is closed in $\mathbb{R}^2$. Indeed any conic combination reduces to at most two independent normals: for dependent vectors, subtract a scalar multiple of a nonzero linear dependence from the non-negative coefficients until one coefficient vanishes, preserving all remaining non-negativity. Iteration terminates with independence. For a convergent sequence of conic combinations, choose a subsequence using the same independent normals, since there are finitely many choices. Their coefficients are continuous linear functions of the combination and hence have non-negative limits. The limit lies in K_I, proving closedness.

If $w = -\nabla L(\theta_*)$ were outside K_I, let k_0 be its nearest point in that closed convex cone and set $d = w - k_0 \neq 0$. Existence follows by minimising distance on a sufficiently large compact ball. Projection optimality gives $d^\top(k - k_0) \leq 0$ for every $k \in K_I$. Taking $k = 0, 2k_0$ gives $d^\top k_0 = 0$, hence $d^\top k \leq 0$ for every $k \in K_I$. Thus $A_I d \leq 0$, but $\nabla L(\theta_*)^\top d = -w^\top d = -\|d\|^2 < 0$, contradicting the directional inequality. Therefore $w \in K_I$ and has non-negative conic coefficients. These are the active multipliers satisfying $\nabla L + A_I^\top \lambda_I = 0$; inactive multipliers are zero. This proves the polyhedral Farkas argument here and yields (6)–(7). Conversely, for any feasible θ, convexity gives $L(\theta) \geq L(\theta_*) + \nabla L(\theta_*)^\top(\theta - \theta_*)$. Stationarity and complementary slackness make the latter inner product non-negative, proving sufficiency. No strict feasible point is used.

At $B = 0$, positivity of p makes zero the only feasible point. Set upper multipliers to zero and choose $\lambda \geq \max\{0, 2h_s/S, 2h_c/C_K\}$; then lower multipliers $\ell = \lambda p - 2h$ are non-negative and all KKT equations hold. For $B > 0$, sufficiently small positive positions also demonstrate strict feasibility, but necessity does not depend on that observation.

For the exact candidate method, an interior optimum has $\theta = M^{-1}h$. A relative interior point of any nondegenerate face lies on one of the five supporting lines. Writing that line as $x_0 + tv$, $v \neq 0$, its restricted objective has positive second derivative $2v^\top M v$, and its unique stationary parameter is

$$t_* = \frac{v^\top(h - Mx_0)}{v^\top M v}. \tag{18}$$

Keep this candidate only if feasible. Every vertex is an intersection of two independent active line normals. There are at most ten line pairs; parallel pairs have no unique intersection. These alternatives cover an optimum on every face, including the singleton at $B = 0$. Testing at most $1 + 5 + 10 = 16$ candidates therefore finds the global minimum; duplicate candidates and empty faces can be discarded.

A.4 Why a variance control is a different financial hedge

Expanding means and covariances gives $M = G + mm^\top$ and $h = k + me$, hence

$$L(\theta) = \mathrm{Var}(Y - \theta^\top g) + (e - m^\top\theta)^2. \tag{19}$$

The variance minimiser is $\theta_v = G^{-1}k$. The identity $(G + mm^\top)G^{-1}m = (1 + \zeta)m$ shows directly that $\theta_v + G^{-1}m\eta/(1 + \zeta)$ solves $M\theta = h$. Uniqueness proves (8); substituting the difference into

(17) gives $\eta^2(\zeta+\zeta^2)/(1+\zeta)^2 = \eta^2\zeta/(1+\zeta)$. Under $P = Q$, fair discounted instrument gains and claim premia have zero means, so $m = e = 0$. If the measures differ, equality can still hold when $\eta = 0$; a strict discrepancy is not automatic.

A.5 Budget value, shadow prices and local sensitivity

Feasible-set inclusion makes v non-increasing. For $B_1, B_2 \geq 0$, a convex combination of their optimisers is feasible at the same convex combination of budgets. Convexity of L then gives $v(tB_1 + (1-t)B_2) \leq tv(B_1) + (1-t)v(B_2)$ for $0 \leq t \leq 1$. This proves convexity of the value function.

For any optimiser at B and its KKT multipliers, the box multipliers give non-negative terms against any other box-feasible point. The budget term gives $L(\theta) \geq v(B) - \lambda_B(p^\top\theta - p^\top\theta_B)$. If $\lambda_B > 0$, complementary slackness gives $p^\top\theta_B = B$; if it is zero, the term vanishes. For $\theta \in \mathcal{F}_{B'}$, the displayed bound is at least $v(B) - \lambda_B(B' - B)$. Minimising proves (9). Thus $-\lambda_B$ is a subgradient on the budget domain, even at a degenerate budget.

For local differentiability, fix an active set I with independent rows, strictly positive active multipliers and strictly inactive remaining constraints. Put $H_0 = 2M$. The matrix

$$\begin{pmatrix} H_0 & A_I^\top \\ A_I & 0 \end{pmatrix} \tag{20}$$

is invertible: a null vector satisfies $H_0x + A_I^\top y = 0$, $A_Ix = 0$, so $x^\top H_0 x = 0$, hence $x = 0$ and then $y = 0$. The KKT solution is consequently affine in B. Strict inequalities persist on an open neighbourhood, making that affine solution the unique optimum there. If I is nonempty, let $\dot{b}_I$ be the vector with one at the budget row when present and zero elsewhere. Then

$$\begin{aligned} \dot\theta &= H_0^{-1}A_I^\top(A_IH_0^{-1}A_I^\top)^{-1}\dot b_I, \\ v' &= -\lambda_B, \qquad v'' = \dot\theta^\top H_0\dot\theta \geq 0. \end{aligned} \tag{21}$$

The first identity follows by differentiating the KKT system; the second uses $\nabla L^\top\dot\theta = -\lambda_I^\top\dot b_I$. For an empty active set, $\theta = \theta_M$ is locally constant. At active-set changes or dependent rows, this differentiability argument is not asserted; the global subgradient bound still holds.

In the budget-only active regime, put $z_p = p^\top M^{-1}p > 0$. Solving stationarity and $p^\top\theta = B$ explicitly gives

$$\begin{aligned} \theta &= \theta_M - \frac{p^\top\theta_M - B}{z_p}M^{-1}p, \\ \lambda_B &= \frac{2(p^\top\theta_M - B)}{z_p}, \\ v(B) &= L(\theta_M) + \frac{(p^\top\theta_M - B)^2}{z_p}. \end{aligned} \tag{22}$$

These formulas apply only while the box is strictly inactive and $\lambda_B > 0$.

Finally, for sufficiently small $B > 0$, write $\theta = Bz$. The upper bounds are then inactive on $z \geq 0$, $p^\top z \leq 1$, since $z_i \leq 1/p_i$. On this compact set, $(L(Bz) - c)/B = -2h^\top z + Bz^\top Mz$ converges uniformly to $-2h^\top z$. Its minimum therefore tends to $-2\max\{0, h_s/S, h_c/C_K\}$, proving the right derivative at zero. No data-dependent active-set heuristic or estimated multiplier is needed for this exact financial sensitivity.

A.6 A bound from feasible KKT diagnostics

For any exactly feasible θ and non-negative multipliers λ, put $r_* = \nabla L(\theta) + A^\top\lambda$, $\varepsilon_c = \lambda^\top(b - A\theta) \geq 0$ and $m_* = \lambda_{\min}(M) > 0$. With $\delta = \theta_* - \theta$, quadratic expansion and feasibility give

$L(\theta_*) \geq L(\theta) + r_*^\top \delta + m_* \|\delta\|^2 - \varepsilon_c$. Completing the square and using the optimum's variational inequality yield

$$m_* \|\theta - \theta_*\|^2 \leq L(\theta) - L(\theta_*) \leq \varepsilon_c + \frac{\|r_*\|^2}{4m_*}. \tag{23}$$

The left inequality follows by expanding at θ_* and using $\nabla L(\theta_*)^\top (\theta - \theta_*) \geq 0$. Floating-point feasibility, residuals and eigenvalues do not themselves prove the exact premises or certify this bound.

References

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